I Year I sem Regular JAN/FEB 2024 AN

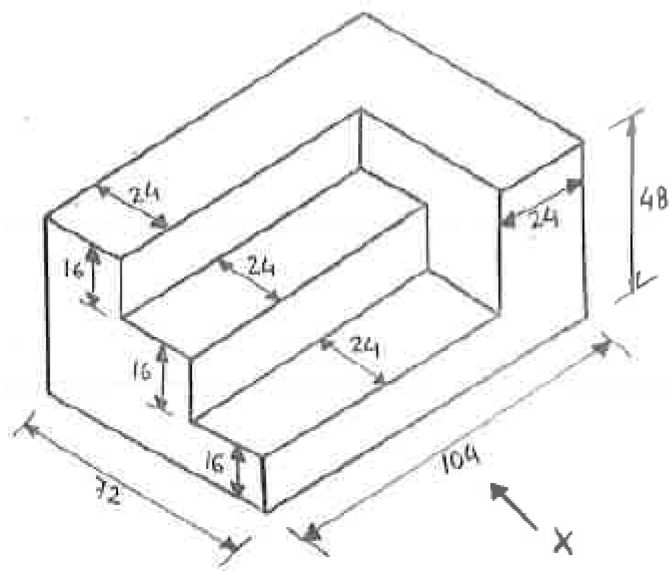
A. Define Hyperbola.
B. The curve generated by a point on the circumference of a circle, which rolls without slipping along outside of another circle is known as
C. When a line is perpendicular to H.P. Can we get its true length in Front view or top view?
D. The top view of a hexagonal plane, when its surface is parallel to V.P. and perpendicular to H.P. is
E. If a cone is resting on one of its generators on H.P., then top view of its base will be seen as a) ELLIPSE (6) CIRCLE
F. A right circular cone resting on HP on its base is cut by a section plane inclined to H.P and bisecting its axis. The true shape of section is an (a)ELLIPSE (b) CIRCLE
G. The development of the lateral surface of a pentagonal pyramid isa) five rectangles b) five triangles.
H. Name the method used for obtaining the developments of pyramids and cones.
I. Isometric view of a circle is
J. The other name of isometric view is

2. Construct a scale of 1:40 to read meters and decimeters and long enough measure up to 6 meters, Mark a distance of 4.7 m on it.

10M
3. Draw rectangular or equilateral hyperbola, given the position of point P on it, 40 mm and 20mm from the asymptotes OX and OY respectively
10M
4. A line PQ has its End P. 10mm above the HP and 20 mm in front of the VP The end Q is 85 mm in front of the VP. The front view of the line measures 75 mm. The distance between the end projectors is 50 mm. Draw the projections of the line and find its true length.
10M
5. A regular pentagon of 30 mm side is resting on one of its edges on HP which is inclined at 45° to VP. Its surface is inclined at 30° to HP. Draw its projections 10M
6. A cylinder of base 50 mm diameter and axis 60 nim long is resting on the point of its base on HP so that the axis is inclined at 30° with HP. draw the projections of the cylinder when the top view of the axis is inclined at 45° with XY.
10M
OR
7. A square prism having a base with a 45mm side and 65mm long axis has its axis parallel to and 55mm in front of the V.P. An edge of its base is parallel to the H.P. and inclined at 30 in the V.P. Draw its projections 10M

8. Draw the isometric projection of the frustum of a hexagonal pyramid of base side 40 mm, top side 25 mm and height 70 mm. The frustum rests on the base of the HP.
10M
OR
9. A cone, base 50mm diameter and axis 60mm long, rests with its base on HP. A section plane perpendicular to VP and inclined at 45° to HP blinds the axis of the cone. Draw the development of the lateral surface of the remaining portion of the cone.
10M

10. A cylinder of base diameter 40 mm and axis height 60 mm is surmounted over a square slab of 60 mm side and 25 mm thickness. Draw the isometric View of the combination.
10M
OR
11. Figure shows a pictorial view of an object. Draw the a) Front view, b) Top view and c) side view. Dimension (All dimensions are in mm)

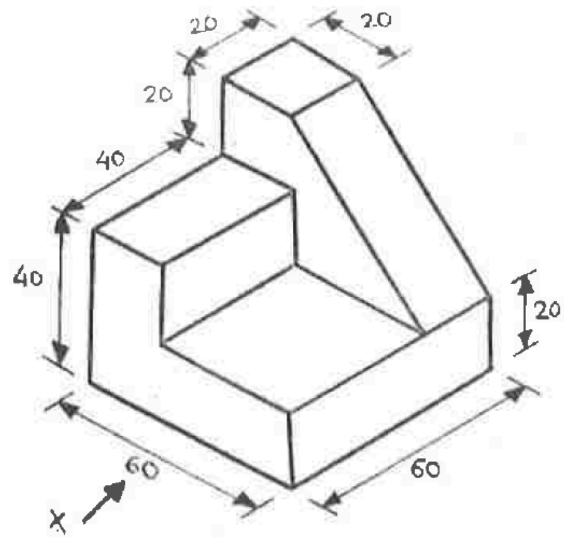


I B. Tech 1 Year 1 Semester (Regular) Final Examinations, Jan/Feb 2024 FN

a. Define the term hyperbola
b. Write the formula for representative fraction
c. When the line is perpendicular to HP, the view from top is_____________
d. A rectangle plane surface is parallel to VP. The true shape is projected on______________
e. Which is the best example of a solid of revolution a) cone Tetrahedron b) pyramid.
f. Shapes of the front view, side view and top view of a square prism standing on HP and one edge perpendicular to VP are ___________ and _______

g. The development of square pyramid is four triangle faces True/false
h. Name the method used for obtaining the developments of pyramids.
i. What is an isometric view.?
j. What is an orthographic view ?

1. Construct a scale of 1 foot 1.5 inch in drawing to read yards and feet and long enough to measure up to 4 feel,
2. Construct an Ellipse when the distance of focus from the directrix is equal to 60 mm and eccentricity is 2/3. Draw a tangent and a normal to the curve at a distance of 70 mm from the directrix.
3. A 90 mm line is parallel to and 25 mm in front of the VP. Its one end is in the HP. While the other is 50 mm above the H.P. Draw its projection and find its inclination. 10M
4. A regular pentagonal plane of 30 mm side is resting on one of its edges on HP which is inclined at 45° to VP. Its surface is inclined at 30° to HP. Draw its projections 10M
5. A cylinder of base 50 mm diameter and axis 60 mm long is resting on a point of its base on HP so that the axis is inclined at 30° with HP. draw the projections of the cylinder when the top view of the axis is inclined at 45° with XY. 10M
6. A Right regular hexagonal pyramid, side of base 30 mm and height 70 mm, rests on its base in HP. With one of its base edges parallel to VP. A horizontal section plane parallel to the HP. And perpendicular to VP cuts the pyramid at a height of 34 mm from the base. Draw its front and sectional top view. 10M
7. Draw the isometric projection of the frustum of a cone of base side 60mm diameter, top side 40 mm diameter and height 70 mm. The frustum rests on the base of the H.P. 10M
8. A Pentagonal prism base 30 mm side and axis 50mm long resting with its base on HP and one edge of the base is parallel to away from VP. It is cut by a section plane perpendicular to VP. And inclined at 50° to the HP. And bisecting the axis. Draw the development. 10M
9. A cone of base diameter 40 mm and axis height 60 mm is surmounted over a square slab of 60 mm side and 25 mm thickness. Draw the isometric view of the combination. 10M
10. Figure shows a pictorial view of an object. Draw the a) Front view, b) Top view and c) side view. Dimension (All dimensions are in mm)



B.Tech I Year I Semester (Supply) End Examinations FEB-2024

A. Define eccentricity and state its values for different conic sections.
B. Define plain scale.
C. When a line is perpendicular to H.P. Can we get its true length in Front view or top view?
D. A rectangle plane surface is parallel to VP. The true shape is projected on
E. Define the terms i) prism and ii) pyramid
F. Classify the different types of solids.
G. The development of lateral surface of a pentagonal pyramid is
   a) five rectangles
   b) five triangles.
H. Name the method used for obtaining the developments of pyramids.
I. List the two advantages of isometric projections.
J. What is the principle followed in the box method of obtaining the isometric projection of an object?

2. Construct a scale of 1:5 to show decimeters and centimeters and long enough to measure up to Im. Show a distance of 6.3 dm on it. 10M

3. Draw an epicycloid having a generating circle of diameter 75 mm and a directing curve of diameter 200 mm. Also draw a normal and a tangent at a point P on the curve. 10M

4. A line PQ has its End P 10mm above the HP and 20 mm in front of the VP. The end Q is 85 mm in front of the VP. The front view of the line measures 75 mm. The distance between the end projectors is 50 mm. Draw the projections of the line and find its true length. 10M

5. A regular pentagonal plane of 30 mm side is resting on one of its edges on HP which is inclined at 45° to VP. Its surface is inclined at 30° to HP. Draw its projections. 10M

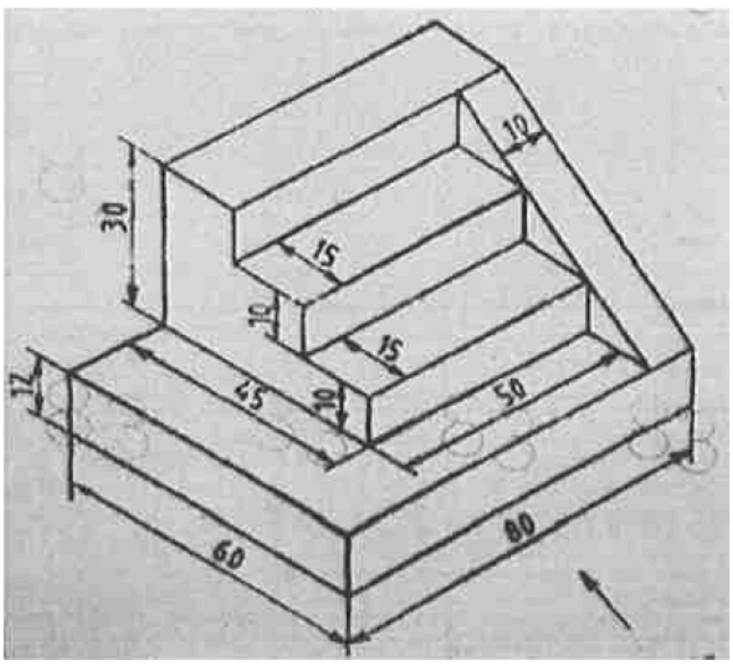
6. A square pyramid, having a base with a 40 mm side and 60 mm axis, is resting on its base on the HP., Draw its projections when (a) a side of the base is parallel to the VP., (b) a side of the base is inclined at 30° to the VP., and (c) all the sides of the base are equally inclined to VP. 10M

7. A cylinder with a 50 mm base diameter and a 65 mm long axis, has a generator in the V.P. and is inclined at 45º to the H.P. Draw its projections. 10M

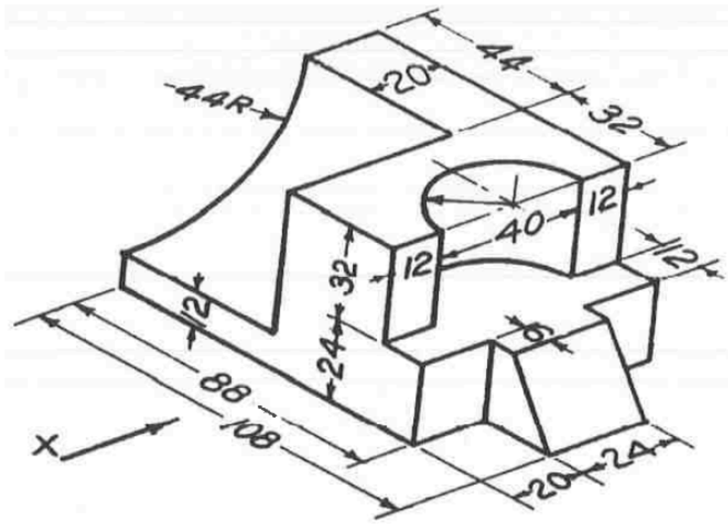
8. A cylinder with 60 mm base diameter and 70 mm long axis, is resting on the ground with its axis vertical. A section plane inclined at 45° to the H.P cuts the cylinder such that the plane passes through the top of the generator and bisecting the axis. Draw the development of its lateral surface. 10M

9. A Pentagonal prism base 20 mm side and axis 40mm long resting with its base on HP and one edge of the base is parallel to away from VP, It is cut by a section plane perpendicular to VP. And inclined at 50 to the HP. And bisecting the axis. Draw the development. 10M

10. Draw the front view, top view, and side view of the given figure. (Note: All dimensions are in 'mm')



11. Draw the front view, top view, and side view of the given figure. (Note: All dimensions are in 'mm')



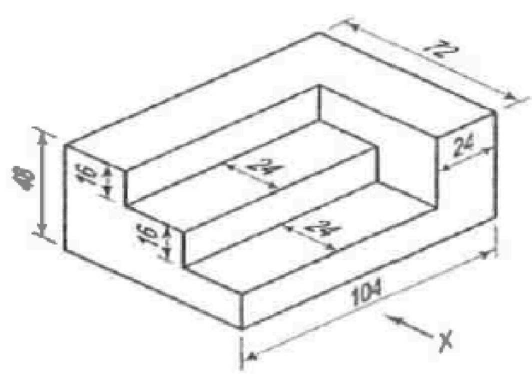
B. Tech I Year I Semester (Supply) End Examinations, March -2024 (Special Batch)

a. Explain various lines used in engineering graphics.
b. Draw a regular hexagon of base 50mm.
c. Draw the projections of the point E-15 mm above the H.P. and 50 mm behind the V.P.
d. A 70 mm. the line PQ is parallel to both H.P. and V.P. One end of the line is 30mm in front of V.P. and 20 mm above the H.P.
e. Draw the projections of a cube of 50mm side, resting on HP such that its vertical faces are equally inclined to VP.
f. Define cylinder and cone in terms of surface of revolution
g. List the classification of Surfaces in development of surfaces.
h. Differentiate between Parallel line and Radial line methods.
i. List the steps involved to convert Isometric drawing into orthographic.
j. Draw the isometric view of a circular lamina of 50mm diameter on any two principle plans using four centers method.

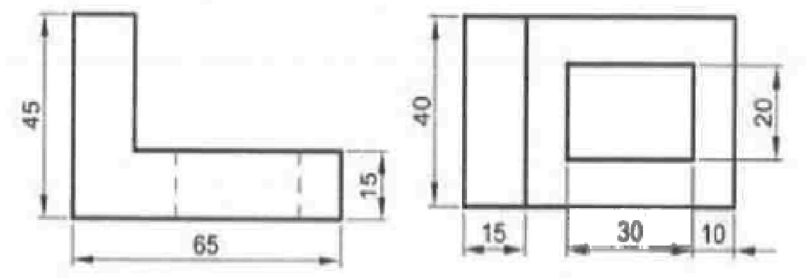
2. Draw a hyperbola when the eccentricity is 3/2 and the distance between focus and directrix is 55mm. Also draw tangent and normal to the curve at a point 40mm from focus. 10M

3. Draw an epi-cycloid of a circle of diameter 50 mm, which rolls outside a circle of diameter 150mm for one revolution. Also draw a tangent and a normal to the epicycloids at a point 110 mm from the center of the directing circle? 10M
4. Draw the projections of a straight line AB of 100 mm long when one of its ends is touching the VP and the other end touching H.P. The angles of inclination with HP and VP are 40° and 50° respectively.   10M
5. A regular pentagon of 30mm side has one side on the ground and when a side is perpendicular to V.P. its plane is inclined at 45º to H.P. Draw the projections.   10M
6. Differentiate between a prism and a pyramid, draw its projections representing front view and top view.   10M
7.   A cone of base 50 mm diameter and axis 65 mm long rests with its base on H.P .It is cut by a section plane perpendicular to V.P, inclined at 450 to H.P .And passing through a point on the axis 35 mm above the base. Draw the sectional top view and true shape of the section. 10M
8. Draw the development of lateral surface of a hexagonal prism with side of base 30 mm long and axis 60 mm long, resting on one of its rectangular faces on the H.P. Its axis is perpendicular to the V.P.   10M

9.   A hexagonal prism, having a base with a 20mm side and 60mm height is resting on the base in HP such that one of the rectangular faces is parallel to the VP. It is cut by a plane perpendicular to VP and 60 degrees inclined to HP and cutting the midpoint of the axis of the solid. Draw development of the lateral surface of the bottom part of the solid.   10M
10. Draw the orthographic projections of the below figure: 10M



11.The front and top views of the angle plate shown in Fig. Draw its isometric view.



B.Tech I Year II Semester (Regular) End Examinations, March 2024 (Special Batch)

a. How does dimensioning contribute to the clarity of engineering drawings?
b. What are the different types of scales used?
c. Explain the auxiliary plane method in projecting planes of geometrical shapes.
d. A hexagonal plane is perpendicular to HP and VP. Draw projection and show true shape inside view.
e. What is the importance of projection in engineering
f. What is the difference between auxiliary view and sectional view
g. How can CAD help in surface development
h. Explain how a funnel is made.
i. Distinguish clearly between isometric view and isometric projection.
j. Brief the Principles of Isometric Projection

2. Draw an epicycloid having a circle of diameter 75 mm and a directing curve of radius 200 mm. Also draw a normal and a tangent at a point P on the curve. 10M

3. The distance between city A and City B is 15km. On inspection of the road map, its equivalent distance measures 3cm. Draw a diagonal scale to read 50m minimum. Show a distance of 3.85km on it. 10M

4. The end A of a line AB is in the H.P. and 25 mm behind the V.P. The end B is in the V.P. and 50 mm above the H.P. The distance between the end projectors is 75 mm. Draw the projections of AB and determine its true length, traces and inclinations with the two planes. 10M

5. An isosceles triangle PQR having the base PQ 50 mm long and altitude 75 mm has its corners P, Q, and R 25 mm, 50 mm and 75 mm respectively above the ground. Draw its projections. 10M

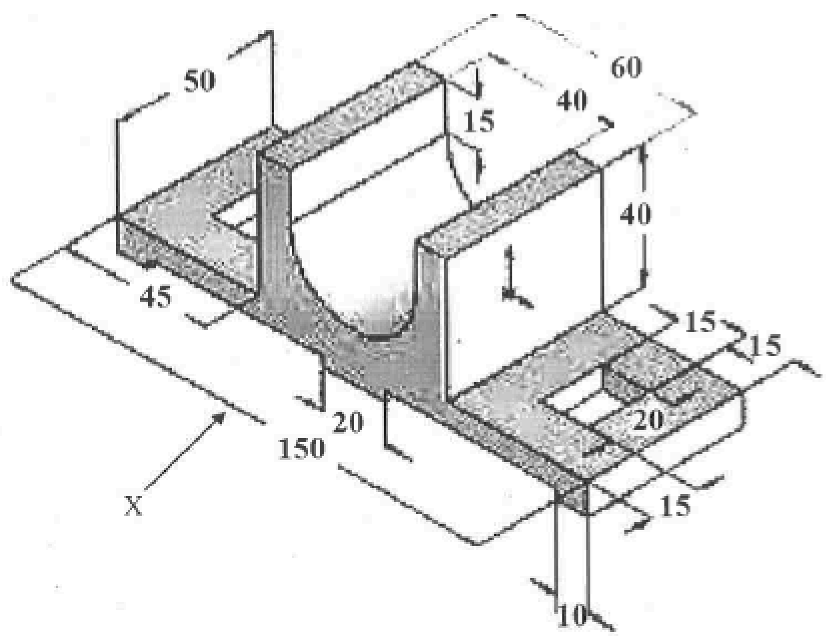
6. A square pyramid, base 35mm side and axis 85mm long, has a triangular face on the ground and the vertical plane containing the axis makes an angle of 45° with the VP. Draw its projections. 10M

7. A Tetrahedron of 75 mm long edges has one edge parallel to the H.P. and inclined at an angle of 45" to the V.P. While a face containing that edge is vertical. Draw its projections.
10M

8. A cone made up of Aluminium sheet with base circle diameter 65 mm and axis length 75 mm is kept with its base on the ground. A circular hole of 30 mm diameter is cut through the cone such that its axis remains perpendicular to V.P.; 10 mm to the right of the axis of cone and 25 mm above the base of cone. Develop the surface of the cone. 10M

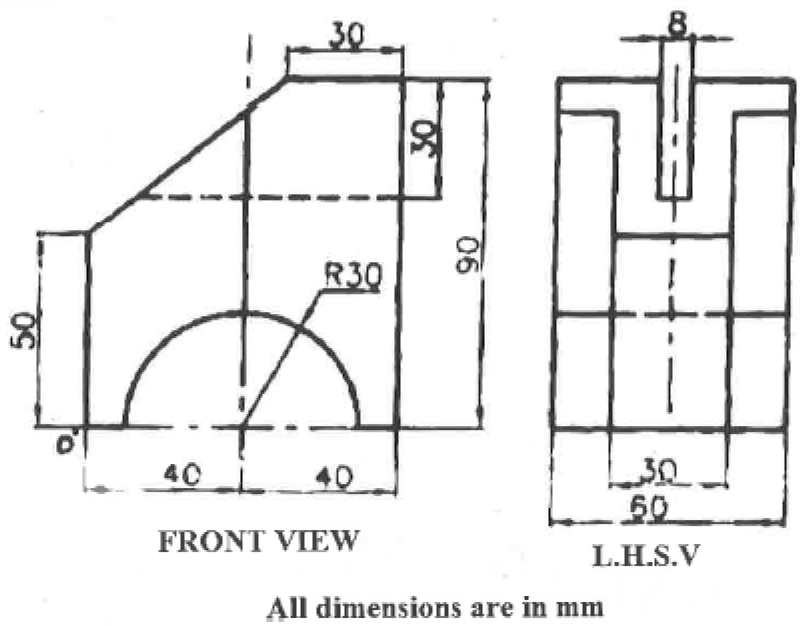
9. A cylinder of diameter 50 mm and height 75 mm is resting on the ground on its flat end. It is cut by a sectional plane inclined at 30° to the axis of the cylinder and passing through a point on the axis at a height of 50 mm from the base. Draw the lateral surface of the bottom part. 10M

10. The isometric view is shown in the figure below. Draw the front view, top view and side view (looking from right). All dimensions are in mm.



10M

11. Draw isometric projection of the object shown in figure, using natural scale.



10M

B.Tech I Year II Semester (Supply) End Examinations FEB-2024

a. Define Hyperbola.
b. Write the formula for representative fraction
c. What is a projection? How is the projection of an object obtained
d. Sketch the symbols used to represent (i) first angle projection and (ii) Third angle projection.
e. If a cone is resting on one of its generators on H.P., then top view of its base will be seen as ELLIPSE (b) CIRCLE
f. Shapes of the front view, side view and top view of a square prism standing on HP and one edge perpendicular to VP are--- and
g. What are the uses of the development of surfaces?
h. What are the different methods of development of surfaces in engineering drawing?
i. Isometric view of a circle is
j. What is an orthographic view?

2. Construct a scale of 1:40 to read meters and decimeters and long enough to measure up to 6 meters. Mark a distance of 4.7 m on it. 10M

3. Construct an Ellipse when the distance of focus from the directrix is equal to 60 mm and eccentricity is 2/3 Draw a tangent and a normal to the curve at a distance of 70 mm from the directrix. 10M

4. A Line AB, 65 mm long, has its end A 20mm above the HP and 25mm in front of the VP. The end B is 40mm above the HP and 65 mm in front of the VP. Draw the projections of AB and show its inclinations with the HP and the VP 10M

5. A pentagonal plane with a 30 mm side rests on the H.P. on an edge such that the surface is inclined at 45^{o} to the H.P. and the edge on which it rests is inclined at 30^{0} to the V.P. Draw its Projections. 10M

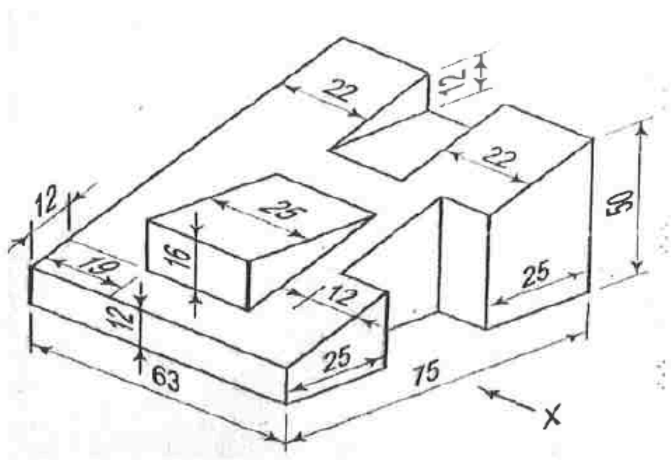
6. A cylinder of base 30 mm diameter and axis 45 mm long is resting on a point of its base on HP so that the axis is inclined at 30 deg with HPdraw . the projections of the cylinder when the top view of the axis is inclined at 45^{o} with XY. 10M

7. A triangular prism, having a base with a 50 mm side and an 80 mm long axis, is lying on one of its rectangular faces on the HP. with its axis perpendicular to the HP., It is cut by a section plane parallel to and 20 mm above the HP., Draw its front view and sectional top view. 10M

8. A cone, base 50mm diameter and axis 60mm long, rests with its base on HP. A section plane perpendicular to VP and inclined at 45^{o} to HP bisects the axis of the cone. Draw the development of the lateral surface of the remaining portion of the cone. 10M

9, A Pentagonal prism base 20 mm side and axis 40mm long resting with its base on HP and one edge of the base is parallel to away from VP. It is cut by a section plane perpendicular to VP. And inclined at 50^{o} to the HP. And bisecting the axis. Draw the development.

10. Draw the a) front view b) side view from the left c) Top view as shown in the figure. (Note: All dimensions are in 'mm')



11. A cone of base diameter 40 mm and axis height 60 mm is surmounted over a square slab of 60 mm side and 25 mm thickness. Draw the isometric view of the combination.

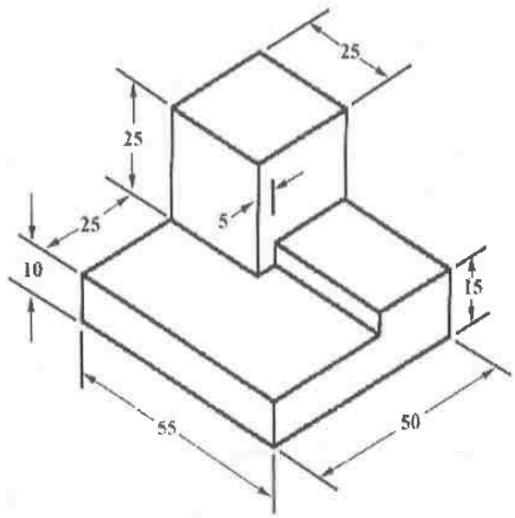
B.Tech I Year I Semester (Regular) End Examinations, October-2023

a. Inscribe a hexagon in a circle of diameter 70 mm
b. Define RF and calculate RF to construct a scale of 1:4 to show cms and long enough to measure up to 5 decimeters.
c. Draw the projections of a point P lying 40mm above HP and 30mm behind VP
d. Why are the second and fourth angles of projections not followed?
e. Recall Polyhedra. Enlist types of polyhedral.
f. Define the terms i) prism and ii) pyramid.
g. Define development of a surface of a solid.
h. Recall any 2 different developments of surface methods?
i. Draw the isometric view of a circle of diameter 60 mm.
j. What is the necessity of isometric projection?

2. Draw regular square, hexagon, octagon of base 70mm. 10M
3. Construct an ellipse whose focus is 70 mm away from the directrix and eccentricity is 2/3.
A point P is 50 mm from both the reference planes. Draw the projections in all possible positions. 4M

Line AB 70mm long and is inclined to HP. If end A is 15mm in front of VP & its front view measures 50mm length, draw its projections and find its inclination with VP. 6M

5. A square ABCD of 40mm side has a comer on HP and 20mm in front of VP. All the sides of the squares are equally inclined to the HP and parallel to the VP. Draw its projections. 10M
6. Explain different types of prisms and pyramids. 10M
7. A cylinder of base diameter 50 mm and axis 80 mm long is resting on HP on one of its generators with the axis parallel to both HP and VP. Draw its projections. 10M
8. Draw the development of the surface of a Pentagonal pyramid of base 30 mm and axis 60 mm having its base on HP and an edge of the base parallel to VP. 10 M
9. A cone of base radius 30 mm and axis 70 mm has its development of its lateral surface as a sector of the circle. What is the radius of the sector and the angle of the sector? 10M
10 Draw the isometric view of a square prism of base 35 mm sides and height 60 mm. 10M
11. Draw the front, top and side views of the given solid.



B.Tech I Year I Semester (Regular) End Examinations, October 2023

a. Define eccentricity and give its significance in deciding different conic sections,
b. Enlist different types of lines that are commonly used in engineering drawing.
c. A point A is 20mm below H.P and 30 mm behind V.P. Draw its projections.
d. What is a projection? How is the projection of an object obtained?
e. How is the true shape of a section obtained?
f. Classify the different types of solids.
g. Why is surface development required?
h. Enlist the steps to be followed for the development of a cylinder?
i. What are the advantages of isometric projections?
j. What is an orthographic view? What is its use?

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2. Construct a scale of 1:4 to show centimeters and long enough to measure upto 5 decimeters. 10 M

3. Draw an ellipse by concentric circles method and find the length of the minor axis if major axis 90 mm and distance between foçi is 70mm 10M

4. Mark the projections of the following points on the same common reference line 25mm apart: Point A 40mm in front of VP and 30mm above HP, Point B 40mm below the HP and in VP. 4M

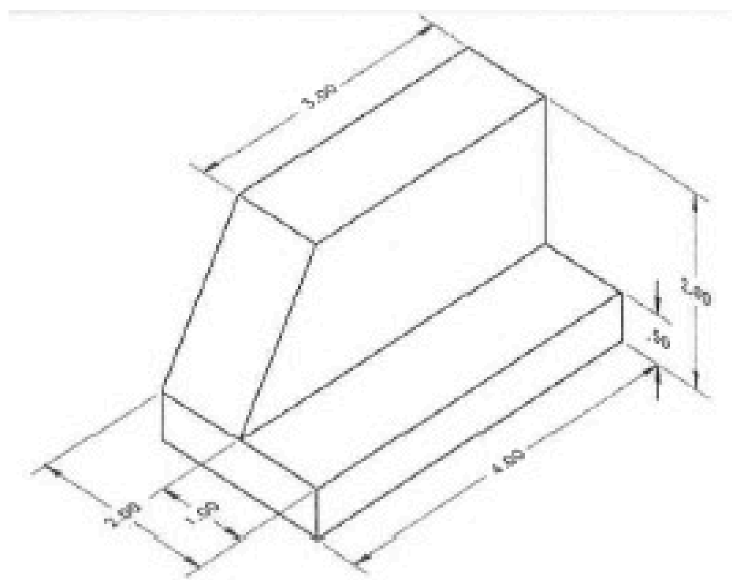
b. A line of 120 mm long is parallel to & 45 mm above HP. Its 2 ends are 30 mm and 60 min in front of VP respectively. Draw its projections and find its inclinations with the VP. 6M

5. A regular hexagonal lamina of 30 mm side rests on one of its sides on HP. It is parallel to and 16 mm away from VP and it is in the first quadrant. Draw its projections. 10M
6. Draw the projections of (a) a cylinder 40mm diameter and axis 50mm long and (b) a cone of base 35mm diameter and axis 60mm long resting on HP with their respective bases.. 10M

7. Explain various types of solids in detail. 10M

8. Draw the development of the lateral surface of a cylinder base 30mm diameter and axis 70mm long, resting on its base on the HP. 10M
9. Develop the lateral surface of a Hexagonal prism of base edge 30 mm and axis 60 mm, resting on one of its rectangular faces on the HP and axis perpendicular to VP. 10M
10 Sketch the isometric view of a square pyramid of 30 mm sides and 75 mm axis resting on its base on ground. 10M

11. Draw the front, top and side views of the given solid.



B.Tech I Year I Semester (Regular) End Examinations, August-2023

a. Inscribe a regular heptagon in a circle of 60 mm diameter.
b. Recall Representative fraction.
c. Reproduce the sketch of all four quadrants in orthographic projections,
d. Summarize: Horizontal plane, Vertical plane and Profile plane.
e. Recall Solids of Revolution. Enlist types of Solids of revolution.
f. Distinguish between a prism and pyramid.

g. Formulate the equation for angle of a sector in development of a cone.
h. Enlist various methods for development of surfaces.
i. Illustrate isometric axes, isometric line and isometric plane.
j. Enlist various methods adopted for drawing isometric views.

2. Draw a regular Pentagon, Heptagon and Octagon of base 50mm. 10M

3.Draw an ellipse by concentric circles method and find the length of the minor axis if major axis = 100 mm and distance between foci is 80mm. 10 M

4. A point P is 65 mm from both the reference planes. Draw the projections in all possible positions. 4M
b. Explain Quadrant theory in detail by taking point as an example. 6M

5. A regular hexagonal lamina of 20 mm side rests on one of its sides on HP. It is parallel to and 12 mm away from VP and it is in first quadrant. Draw its projections. 10M

6. A hexagonal prism has one of its regular faces parallel to the HP. Its axis is perpendicular to VP and 35mm above the ground. Draw its projections when the nearer end is 20mm in front of the VP. Side of base 25mm long; axis 50mm long. 10M

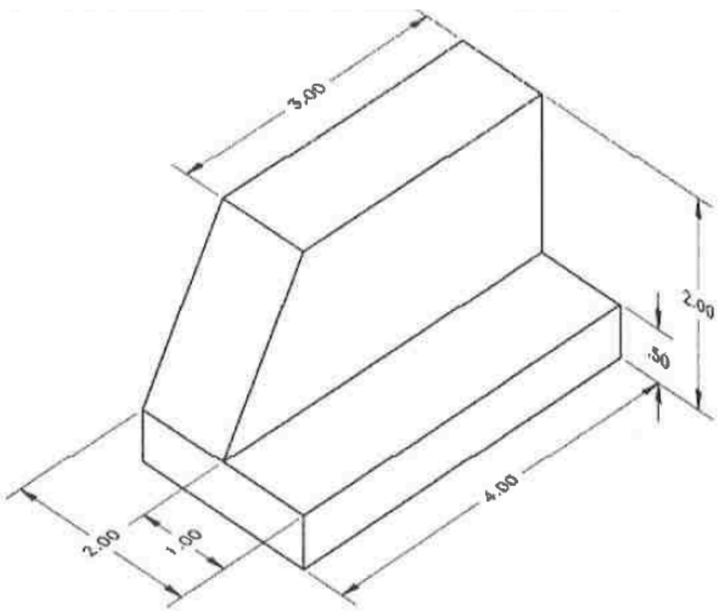
7. A right regular pentagonal pyramid, edge of base 30 mm and 68mm high, is lying on one of its triangular faces on ground plane (HP) such that its axis is parallel to VP. Draw its projections. 10M

8. Draw the development of a right circular cone of base 40 mm diameter and axis 60 mm 10M

9. Develop the lateral surface of a square-base prism of base edge 20 mm and axis 40 mm 10M

10. Sketch the isometric view of a square pyramid of 40 mm sides and 75 mm axis when its axis is horizontal 10M

11. Draw the front, top and side views of the given solid.



B.Tech I Year I Semester (Regular) End Examinations, October-2023

a. Inscribe a pentagon in a circle of diameter 60 mm
b. Enlist types of scales used in Engineering drawing
c. Recall orthographic projections
d. Sketch the projections of a point 2 cm below the HP and 4 cm behind the VP
e. Recall Polyhedra. Enlist types of polyhedra
f. Distinguish between a oblique solid and right solid
g. Define development of a surface of a solid
h. Enlist applications of development of surfaces
i. What is meant by isometric drawing? What are its advantages
j. Formulate the relation between true length and isometric length based on isometric scale

2. Construct a scale of 1:60 to show metres and decimetres and long enough to measure upto 6 metres 10M

3. Construct an ellipse whose focus is 50 mm away from directrix and eccentricity is 2/3. 4M
Distinguish between first angle and third angle projections 6M

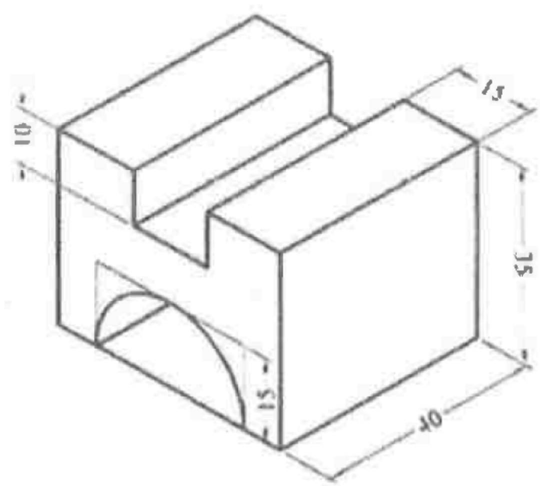
6. A line of 100 mm long is parallel to & 30 mm above HP. Its 2 ends are 25 mm and 50 mm in front of VP respectively. Find its inclinations with the VP. 10M

5. A regular hexagon of 25 mm side has its one edge on the HP. The surface of the plane is perpendicular to the VP and inclined at 40° to the HP. Draw the three views of the plane 10M

6. Draw projections of a pentagonal prism, edge of base 30mm and 50mm long, having one of its base edges perpendicular to VP. 10M

7. A cylinder of base diameter 40 mm and axis 70 mm long is resting on HP on one of its generators with the axis parallel to both HP and VP. Draw its projections. 10M
8. Draw the development of surface of a Hexagonal pyramid of base 40 mm and axis 60 mm 10M
9. A cone of base radius 40 mm and height 60 mm has its development of its lateral surface as a sector of the circle. What are the radius of the sector and the angle of the sector 10M
10. Draw the isometric view of a square prism of base 40 mm sides and height 70 mm when its axis is vertical 10M

11. Draw the front, top and side views of the given solid 10M



B. Tech I Year II Semester (Supplementary) End Examinations Feb-2024

a. What is meant by eccentricity? Give it values for ellipse, parabola.
b. What is meant by representative fraction?
c. What is the difference between first angle and third angle projection?
d. Draw the projection of a point 'A'. Which is 30mm above VP and in HP.
e. Differentiate between prism and pyramid.
f. How do you know the true shape of a section?
g. What is meant by the radial line method?
h. How is the development of cone drawn?
i. How is an isometric scale constructed?
j. What is the principle of isometric projection?

2. Draw a hyperbola when the distance its focus and directrix is 50mm and eccentricity is 3/2.

10M

3. Construct an Ellipse when the distance of focus from the directrix is equal to 60 mm and eccentricity is 2/3. Draw a tangent and a normal to the curve at a distance of 80 mm from the directrix. 10M

4. A line AB of 70mm long one of its end A is 20 mm above HP and 35mm in front VP. Draw its projections when the line is inclined at 30° IIP and 45º to VP. 10M

5. Draw the projections of a thin square plate ABCD of side 50mm, parallel to VP when its edge AB is on the HP and 25mm in front of VP

6. A hexagonal pyramid, base 25mm side and axis 50mm long has an edge of its base on the ground. Its axis is inclined at 30° to the ground and parallel to the VP. Draw its projections. 10M

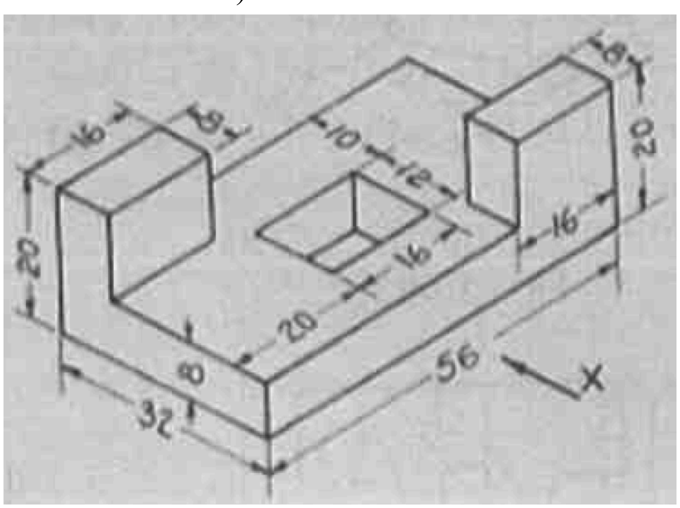
7. A hexagonal pyramid of base side 30 mm and height 70 mm long is resting on its base on the HP with two edges parallel to VP. It is cut by a section plane perpendicular to the VP inclined at 45° to HP and intersecting the axis at a point 25 mm above the base. Draw sectional top view and front view. 10M

8. A square pyramid side of base 40 mm and height 60 mm stands on the HP with a base edge parallel to the VP. It is cut by a plane inclined at 45" to the HP perpendicular to the VP and bisecting the axis. Draw the development of a truncated pyramid. 10M

9. A cone of base diameter of 50mm with 70mm height is resting on HP, and it is cut by an AIP at 45° to HP and bisecting the axis. Draw the development of the remaining portion. 10M.

10. Draw the isometric view of the square prism by taking the side of base as 20 mm long and the axis 40 mm long, when its axis is (i) vertical (ii) horizontal 10M

11. Draw the front view, top view and left side view of the following isometric drawing (all dimensions in mm)



B.Tech I Year II Semester (Supplementary) Examinations, March 2023

a. What are the different types of lines that are commonly used in engineering drawing?
b. What do you mean by Representative Faction (R.F) or Scale Factor (S.F.) 7
c. A pentagonal plane of side 30 mm is resting in the V.P. and when a side is perpendicular to the H.P. Draw its projections.
d. What is meant by projections? Enlist different types of projections?
e. Name the solid whose projections in any position on to any plane are identical Also name shape of the projections?
f. What is meant by frustum of solids? Draw one example?
g. Explain the difference between the Triangulation method and Approximation method.
h. Explain briefly about any 2 different development methods.
i. Draw the isometric view of a semi-circular lamina of 50mm diameter.
j. Draw the isometric view of a hexagonal plane of 30mm side.

PART-B [MARKS: 50]

2. Construct a parabola, with distance of the focus from the directrix as 50 mm.
Also draw a tangent and normal to the curve at a point 40 mm from the directrix.

3.A line AB, 90 mm long, is inclined at 45° to the H.P. and its top view of line makes an angle of 60° with the V.P. The end A is in the H.P. and 12 mm in front of V.P. Draw its front view and find its true inclination with the V.P.

4. A Hexagonal prism of base side 30 mm and axis 70 mm is resting on a corner of its base on the H.P and the axis inclined at 60° to the H.P and parallel to the V.P Draw its projections?

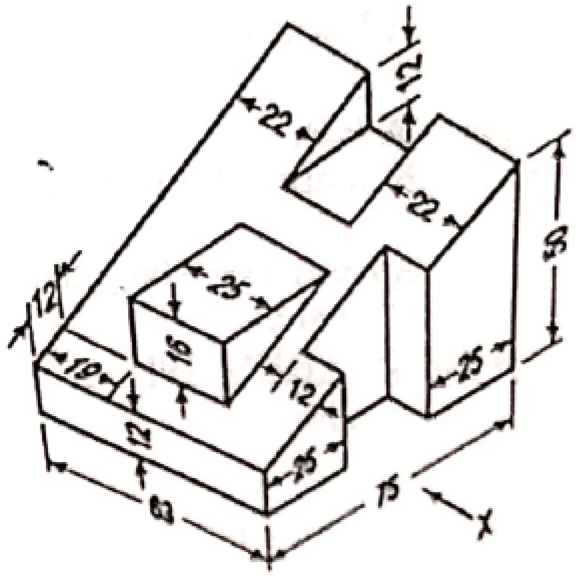
5. A pentagonal pyramid side of base 26 mm and 52 mm height, rests with its base on H.P One of its edges of its base is perpendicular to V.P A section plane perpendicular to V.P and inclined at 45° to H.P, bisect the axis Draw the sectional top view and true shape of the solid?

6. A hexagonal pyramid of base side 30 mm and axis 70 mm is resting on its base on the HP with a base edge parallel to the VP. It is cut by an Auxiliary InclinedPlane (AIP) which is inclined at 30° to the HP and bisecting the axis. Draw the development of the surface of the truncated pyramid.

7. A sphere of diameter 50mm is surmounted centrally on the top of a square block of side 60mm and thickness 20mm. Draw isometric view of arrangement

8. A Semi circular plane of diameter 70 mm has its straight edge on the H.P. the surface makes an angle of 35º to H.P. and the edge makes an angle of 40° to V.P Draw the projections.

9. Draw the three views of given isometric view.



B.Tech I Year I Semester (Supplementary) End Examinations March 2023

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a. Define the terms (i) hyperbola and (ii) parabola
b. Differentiate between a plain and diagonal scale.
c. What is a projection? How is the projection of an object obtained?
d. A point A is 20mm above H.P and 30mm in front of V.P. Draw its projections.
e. Classify the different types of solids.
f. A cone stands on HP with its axis vertical. Show the cutting plane which will produce a circle.
g. What are the different types of development? Explain.
h. Define the term, "line of intersection".
i. Draw the isometric view of a circle of diameter 60 mm.
j. Explain the terms isometric length and true length.

2. Construct a diagonal scale of 1:2.5 showing centimeters and millimeters and long enough to measure up to 20 centimeters. Show 15.4 cm on it.
3. The front and top views of a line AB measures 55 mm and 65 mm respectively. The point A is 10mm above the HP and 15 mm In front of VP. The front view of the line is inclined at 40° to the reference line. Determine the true length of AB and it's inclinations with HP and VP,

4. A hexagonal pyramid, base 25mm side and axis 50mm long has an edge of its base on the ground. Its axis is inclined at 30° to the ground and parallel to the VP. Draw its projections
5. A square pyramid side of base 40 mm and height 60 mm stands on the HP with a base edge parallel to the VP. It is cut by a plane inclined at 45° to the HP perpendicular to the VP and bisecting the axis. Draw the development of a truncated pyramid.
6. Draw the isometric projection of the frustum of a hexagonal pyramid of base side 40 mm, top side 25mm, and height 70mm. The frustum rests on the base of the H.P.
7. Construct an Ellipse when the distance of focus from the directrix is equal to 60 mm and eccentricity is 2/3. Draw a tangent and a normal to the curve at a distance of 80 mm from the directrix.
8. A cylinder of diameter 50mm and height 75mm is resting on the ground on its flat end. It is cut by a sectional plane inclined at 30° to the axis of the cylinder and passing through a point on the axis at a height of 50mm from the base. Draw the lateral surface of the bottom-part.
9. Draw the a) Front view. b) Top view c) right side view. All dimensions are in mm.